



Summer Internship Project Report, 2025

**Spatial Redistribution of Predator-Prey
Dynamics**



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Abstract

This project presents a predator–prey model in a two-patch environment. In this study, we will work with a set of four ordinary differential equations that govern the densities of the prey and predator populations on each patch. The migration rates of prey (respectively, predator) are considered to be dependent on the density of predator (respectively, prey). Here, we assume that migration between patches is faster than predator–prey interactions and other phenomena. Generally, prey leave a patch at a migration rate proportional to the density of predators in that patch. Predators leave a patch at a migration rate inversely proportional to the density of the local prey population in that patch; however, here we consider only a constant migration rate. We utilize two time scales to reduce the complete model to a system of two equations governing the total densities of prey and predators. Firstly, the stability analysis of the aggregated model shows that a unique strictly positive equilibrium exists. This equilibrium may be stable or unstable. Then, we will see that a Hopf bifurcation may occur, leading the equilibrium to be a centre. In this study, the theoretical analysis is followed by numerical simulations using MATLAB to visualize the models by phase portraits and time-series plots.

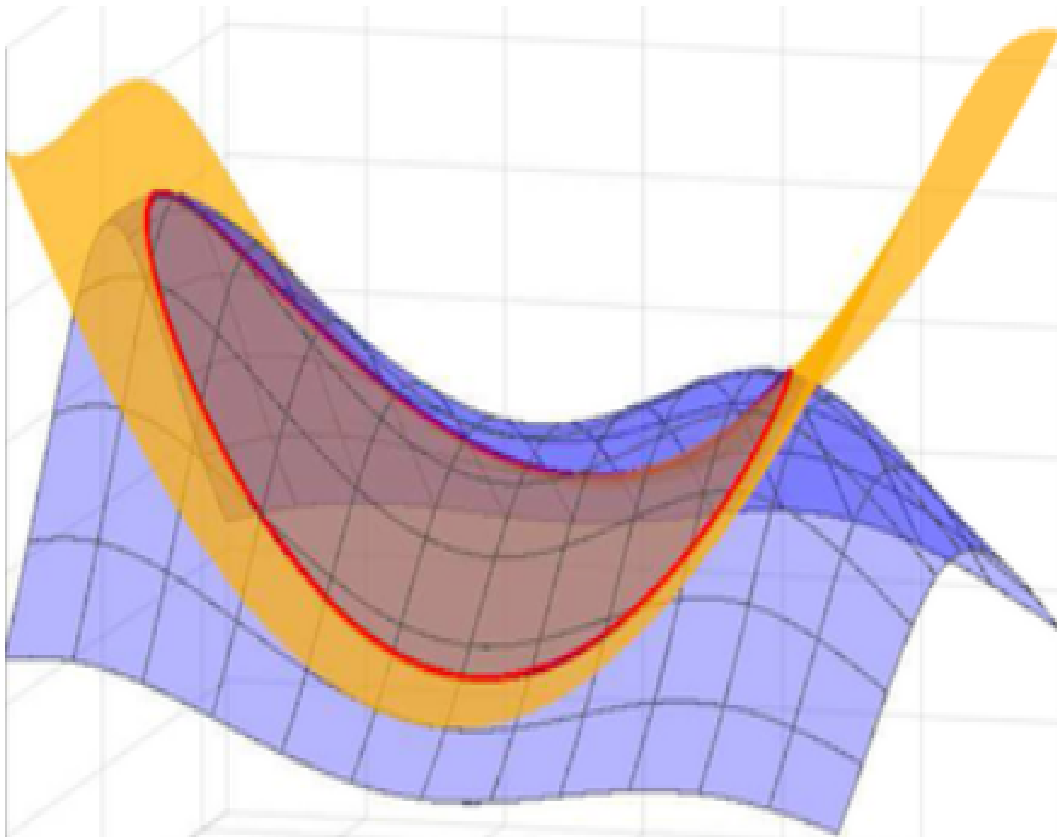
Acknowledgement

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Sincerely,
Arpan Saha

Title of Research Project

Spatial Redistribution of Predator-Prey Dynamics



Objectives

The dynamic interactions between predators and prey is one of the most important areas in mathematical ecology. The spatial redistribution of prey and predators plays a key role in the predator-prey interactions. This internship aims to study exactly the same thing in detail. These studies introduce models that capture nonlinear dispersal mechanisms, explore their stability properties, and highlight the role of spatial structure in ecological systems. The following objectives were pursued during the course of the internship.

Mathematical Interpretations of the Systems

This project is mainly based on the application of systems of differential equations on ecological models. Any ecological model can be described as a set of differential equations whose numerical or analytic solutions reveal the real truth of the same.

To Analyze the Mathematical Models

This project work aims to study the mathematical models behind the ecological phenomena using aggregation of variables, stability & bifurcation analysis, with a strong focus on the key parameters that affect the process.

Numerical Simulations

This work aims to conduct numerical simulations that illustrate the theoretical analysis that has been carried out here. The modelling of theoretical concepts into numerical ones to visualize the complete process is one of the most important objectives of this work.

The objectives outlined provide a structured foundation for the subsequent stages of the study, including model formulation, analytical techniques, and interpretation of simulation results. This sets the stage for a deeper investigation into the complex interplay between the spatial redistribution and population stability in ecological systems.

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Chapter 1

Introduction of Research Topic

Classical predator–prey models, such as the Lotka–Volterra model, can be used in a homogeneous environment. However, in general, the environment is heterogeneous, and this can be represented by using a set of discrete patches connected by migrations. In the simplest situation, a set of two patches is used. So, in that case, the mathematical models have two components, one is the predator–prey interaction, and another is the migrations of these from patch to patch. In this scenario, there is a lot of interest in studying the prey density-dependent migrations of predators and their effect on the predator–prey dynamics. However, few studies have been done regarding the predator density-dependent migrations of prey and predator–prey dynamics. This phenomenon, where prey leave patches with high predator density, is ecologically relevant and observed in species like zooplankton, which perform vertical migrations to reduce predation risks from fishes. In this work, we will also try to study the coupling of the two kinds of density-dependent migrations in the concerned scenario. In such a context, this work draws inspiration from the studies of Mchich et al. (2007) and El Abdllaoui et al. (2007). In this work, we will try to explore both studies using some advanced mathematical techniques such as aggregation of variables, bifurcation & stability analysis and several numerical simulations.

Chapter 2

Theoretical Framework of Topic

In this work, we will study a predator–prey model in a two-patch environment. It is a system of four ordinary differential equations that govern the local densities of the predator and prey populations. We assume that there are two different time scales: one rapid, corresponding to dispersal and one slow, corresponding to prey natural growth, predator natural mortality, and prey–predator interactions. After giving details of the methodology for this work, in Chapter 5 (section 5.1), we present the complete predator–prey model, which consists of a system of four ordinary differential equations which govern the local prey and predator population densities. The existence of two time scales helps us to build a reduced model, called the aggregated model, that we have presented in subsection 5.1.2. This reduced model is composed of two equations. It describes the dynamics of total prey and predator populations over a slow time scale t . Then, in the next part, we have done stability analysis of the system and studied bifurcations and bifurcation parameters. In the last part of Chapter 5, we have given some phase portraits and time series plots to support our theoretical analysis. In Chapters 6 and 7, the conclusion, some future suggestions, and key results of this study have been discussed, concluding with references.

Chapter 3

Profile of Organization

Indian Institute of Technology, Kanpur is one of the premier research institutions, established by the Government of India (1959). From its very inception, IIT Kanpur has been striving to develop itself into an institution of excellence in education and research in consonance with the contemporary and future needs of India. Research at IIT Kanpur is done by small individual-led research groups, and there are also large interdisciplinary project teams. The research environment supports highly creative work done by a small group.

3.1 Institutional Research Environment

Research has always been the center of attention at IIT Kanpur. There is a large talent pool of dedicated researchers. So, there is a great opportunity for collaboration here in this ecosystem. Due to the vibrant research environment, there is regular exposure to new ideas. It helps young researchers to prepare their minds and see a broader picture in their area of interest. Here, the research is supported by a dynamic research administration with a highly non-bureaucratic work culture. The quality & diversity of research carried out here is a measure of that, along with the other parameters. Overall, the dedication of researchers, the work culture and the supportive administration are some of the key factors of the ongoing fundamental & applied research done at IIT Kanpur.

3.2 Research in Mathematics

The Department of Mathematics & Statistics at IIT Kanpur has always striven to succeed in the endeavor of achieving excellence in both teaching and research. Over the years, the department has evolved as one of the premier departments in the country, making a remarkable contribution in adding to the number of well-trained, bright mathematicians and statisticians in the country and abroad. The department conducts high-quality research in the areas of algebra, analysis, number theory, differential equations & numerical analysis, probability & statistics. The department also maintains a vibrant research atmosphere through various initiatives. There is a regular stream of visitors to the department, both for the long term and short term, resulting in collaborative research, interactions, and seminars.

Chapter 4

Methodology Followed

The methodology followed in this project work involves both theoretical analysis and numerical simulations to investigate the effects of spatial movements on the stability of predator–prey systems. The overall work is carried out through the following key steps:

4.1 Study of the Complete Model

Considering various assumptions, the complete models with a set of four ordinary differential equations are formed that govern the prey-predator population densities on each patch. In the first model, predators disperse with constant migration rates, while prey dispersal is predator density-dependent. In the 2nd model, both the cases of prey and predators are studied.

4.2 Formation of Aggregated Model

In this step, the system of four 1st order differential equations has been reduced to a system of two 1st order differential equations with the introduction of a new time scale, describing the total prey and predator populations across both patches. This method is an adapted form of the center manifold theorem. The reduced system preserves the essential dynamics over the new time scale.

4.3 Stability Analysis

Equilibrium points of both the complete and aggregated models have been computed analytically. Linear stability analysis was performed using Jacobian matrices evaluated at equilibrium points and eigenvalue analysis to determine conditions for local stability or instability.

4.4 Numerical Simulations

The theoretical conclusions have been verified through numerical simulations using MATLAB. Results of the numerical simulation have been visualized using phase portraits and time series plots.

Chapter 5

Analysis and Interpretation of Research Project

5.1 Predator Density Dependent Dispersal of Prey and Predator Prey Dynamics

5.1.1 Prey-Predator Complete Model

According to some assumptions on several factors, such as migrations and biotic interactions, the complete system, over the rapid time scale $\tau = \frac{t}{\varepsilon}$, is as follows:

$$\begin{cases} \frac{dn_1}{d\tau} = (m'(p_2)n_2 - m(p_1)n_1) + \varepsilon[r_1n_1 - a_1n_1p_1], \\ \frac{dn_2}{d\tau} = (m(p_1)n_1 - m'(p_2)n_2) + \varepsilon[r_2n_2 - a_2n_2p_2], \\ \frac{dp_1}{d\tau} = (k'p_2 - kp_1) + \varepsilon[-m_1p_1 + b_1n_1p_1], \\ \frac{dp_2}{d\tau} = (kp_1 - k'p_2) + \varepsilon[-m_2p_2 + b_2n_2p_2], \end{cases} \quad (5.1)$$

where, $n_1(t)$ & $n_2(t)$ are prey population densities at time t , in patch 1 & 2 respectively. The predators are represented by their patch population densities $p_1(t)$ and $p_2(t)$ at time t . (r_i ($i = 1, 2$) represents the intrinsic growth rate of the prey population in patch i . Since patches have distinct characteristics, we assume that $r_1 \neq r_2$. a_i , b_i are predation parameters on patch i . m_i is the natural mortality rate of the predator in patch i . k & k' are predator migration rates from patch 1 to patch 2 and inversely. The prey migration rates are predator density dependent as follows:

$$\begin{aligned} m(p_1) &= \alpha p_1 + \alpha_0, \\ m'(p_2) &= \beta p_2 + \alpha_0. \end{aligned} \quad (5.2)$$

Here, α_0 , α and β are positive parameters, ε is a small dimensionless parameter.

5.1.2 The Aggregated Model

The first step to form the aggregate model is to neglect the slow part of the equations and study the rapid dispersal model. First, we calculate the rapid equilibrium. We first define,

$$\begin{aligned} n(t) &= n_1(t) + n_2(t), \\ p(t) &= p_1(t) + p_2(t). \end{aligned} \quad (5.3)$$

to be the total prey (respectively, predator) density, and we set $\varepsilon = 0$ in the system (5.1). After some calculations, we get the following equilibria for the rapid part of the system:

$$\begin{cases} p_1^* = \frac{k'}{k+k'}p, \\ p_2^* = \frac{k}{k+k'}p, \\ n_1^* = \frac{\alpha_0 + \beta\mu_2 p}{2\alpha_0 + \delta p}n, \\ n_2^* = \frac{\alpha_0 + \alpha\mu_1 p}{2\alpha_0 + \delta p}n. \end{cases} \quad (5.4)$$

where $\delta := \alpha\mu_1 + \beta\mu_2$. Also, note that, on the rapid time scale, the total prey ($n(t) = n_1(t) + n_2(t)$) and predator ($p(t) = p_1(t) + p_2(t)$) populations are constant. We substitute the rapid equilibria (5.4) and add equations of the local prey and predator densities, and obtain the following system, when using the slow time scale t ,

$$\begin{cases} \dot{n}(t) = \frac{1}{\delta p + 2\alpha_0}(rn + anp - \bar{b}np^2), \\ \dot{p}(t) = -Mp + \frac{1}{\delta p + 2\alpha_0}(bnp + cnp^2), \end{cases} \quad (5.5)$$

where

$$\begin{cases} r = \alpha_0(r_1 + r_2), \\ a = r_1\beta\mu_2 + r_2\alpha\mu_1 - \alpha_0(a_1\mu_1 + a_2\mu_2), \\ \bar{b} = \mu_1\mu_2(a_1\beta + a_2\alpha), \\ M = m_1\mu_1 + m_2\mu_2, \\ b = \alpha_0(b_1\mu_1 + b_2\mu_2), \\ c = \mu_1\mu_2(b_1\beta + b_2\alpha). \end{cases} \quad (5.6)$$

5.1.3 Stability Analysis

From the system (5.5), we get two equilibrium points:

- $(0,0)$.
- (n^*, p^*) , where

$$\begin{aligned} n^* &= \frac{M(\delta p^* + 2\alpha_0)}{ep^* + b}, \\ p^* &= \frac{a + \sqrt{a^2 + 4rb}}{2\bar{b}}. \end{aligned} \quad (5.7)$$

The Jacobian matrix is as follows:

$$J(n, p) = \begin{bmatrix} \frac{r+ap-\bar{b}p^2}{\delta p+2\alpha_0} & n \frac{(a-2\bar{b}p)(\delta p+2\alpha_0)-\delta(r+ap-\bar{b}p^2)}{(\delta p+2\alpha_0)^2} \\ \frac{bp+cp^2}{\delta p+2\alpha_0} & -M + \frac{n(b+cp)}{\delta p+2\alpha_0} + \frac{c(\delta p+2\alpha_0)-\delta(b+cp)}{(\delta p+2\alpha_0)^2} \end{bmatrix}.$$

For the equilibrium point $(0, 0)$, we have

$$J(0, 0) = \begin{bmatrix} \frac{r}{2\alpha_0} & 0 \\ 0 & -M \end{bmatrix}.$$

The matrix has two real eigenvalues with opposite signs. So, it is a saddle point. For the equilibrium point (n^*, p^*) , we have

$$J(n^*, p^*) = \frac{1}{(b+cp^*)(\delta p^*+2\alpha_0)} \begin{bmatrix} 0 & M(2\alpha_0 a - \bar{b}\delta p^{*2} - 4\bar{b}\alpha_0 p^* - \delta r) \\ p^*(b+cp^*)^2 & p^*M(2\alpha_0 c - \delta b) \end{bmatrix},$$

with a positive determinant C :

$C = \frac{p^*M}{(\delta p^*+2\alpha_0)^2}(\bar{b}\delta p^{*2} + 4\bar{b}\alpha_0 p^* + \delta r - 2\alpha_0 a)$, as $4\bar{b}\alpha_0 p^* - 2\alpha_0 a = 2\alpha_0\sqrt{a^2 + 2r\bar{b}} > 0$. The Jacobian matrix of the equilibrium point (n^*, p^*) has a trace B given by,

$$B := \frac{p^*M(2\alpha_0 c - \delta b)}{(b+cp^*)(\delta p^*+2\alpha_0)}. \quad (5.8)$$

Let B be the trace of the Jacobian matrix $J(n^*, p^*)$, and $\Delta := B^2 - 4C$. So, we have two eigenvalues for the matrix given by:

$\lambda_1 = \frac{B-\sqrt{\Delta}}{2}$ and $\lambda_2 = \frac{B+\sqrt{\Delta}}{2}$ if $\Delta > 0$ or $\lambda_1 = \frac{B-i\sqrt{-\Delta}}{2}$ and $\lambda_2 = \frac{B+i\sqrt{-\Delta}}{2}$ if $\Delta < 0$. So, for the stability of the equilibrium point (n^*, p^*) , two situations occur:

1. Case (i): If $B < 0$, it is a stable equilibrium (focus or node). We have either two negative real eigenvalues (when $\Delta > 0$) or two complex eigenvalues with negative real parts (when $\Delta < 0$), and in both cases, (n^*, p^*) is a stable equilibrium (focus or node).
2. Case (ii): If $B > 0$, it is an unstable equilibrium. We have either two positive real eigenvalues (when $\Delta > 0$) or two complex eigenvalues with positive real parts (when $\Delta < 0$), and in both cases, (n^*, p^*) is an unstable equilibrium.

5.1.4 Bifurcation and Bifurcation Parameter

The term B (given by (5.8)) can be equal to zero, and in this case the linearization of the aggregated model is not sufficient to conclude any information about the dynamics of the model. When B crosses the zero value, a Hopf bifurcation occurs. By application of the Poincaré map, the dynamics is a centre at this instant. The Hopf bifurcation in this case is degenerate as there is no limit cycle associated with it. If we set

$$F(n, p) := \frac{\dot{n}}{n \cdot p} \text{ and } G(n, p) := \frac{\dot{p}}{n \cdot p},$$

then we have,

$$F(n, p) := \frac{1}{\delta p + 2\alpha_0} \left(\frac{r}{p} + a - \bar{b}p \right),$$

$$G(n, p) := -\frac{M}{n} + \frac{1}{\delta p + 2\alpha_0} (b + cp),$$

and so, we get,

$$\frac{\partial F}{\partial n} = 0 \text{ and } \frac{\partial G}{\partial p} = \frac{2\alpha_0 c - \delta b}{(\delta p + 2\alpha_0)^2}.$$

For $B \neq 0$, in the connected domain D , which is defined by the positive quadrant, which is positively invariant as the n and p axes are nullclines, there is no closed orbit by the application of Poincaré-Bendixon theorem. Note that this result is also valid for the ε -perturbation of the system (5.5).

5.1.5 Interpretation of the Bifurcation Parameter near the equilibrium point

The trace B of the Jacobian matrix $J(n, p)$ can be written as:

$$B = \frac{-\alpha_0 M}{D} \Delta_{mig} (b_1 \mu_1 - b_2 \mu_2),$$

where,

$$\Delta_{mig} := (\alpha \mu_1 - \beta \mu_2) p^*,$$

and D is given by,

$$D := (b + c p^*)(\delta p^* + 2\alpha_0),$$

Similar Patches

First, we will study the simplest cases when two patches are similar, i.e. we assume that $\alpha = \beta$ and $b_1 = b_2$ but $\mu_1 \neq \mu_2$. Then the trace becomes

$$B = -\frac{\alpha \alpha_0 b_1 M}{D} (\mu_1 - \mu_2)^2 p^2 < 0,$$

and in this case, the equilibrium point (n^*, p^*) is globally asymptotically stable. If $\mu_1 = \mu_2$, then the equilibrium point (n^*, p^*) is neutrally stable, and there are periodic solutions. So, when the two patches are similar, the positive equilibrium is stable.

Different Patches

Now, let us suppose that $\mu_1 > \mu_2$ and that b_1 is close to b_2 . Then, to have a equilibrium at (n^*, p^*) , we must have $\Delta_{mig} > 0$. Thus, the greater α is to β , the greater the stability of the system.

5.1.6 Numerical Simulations

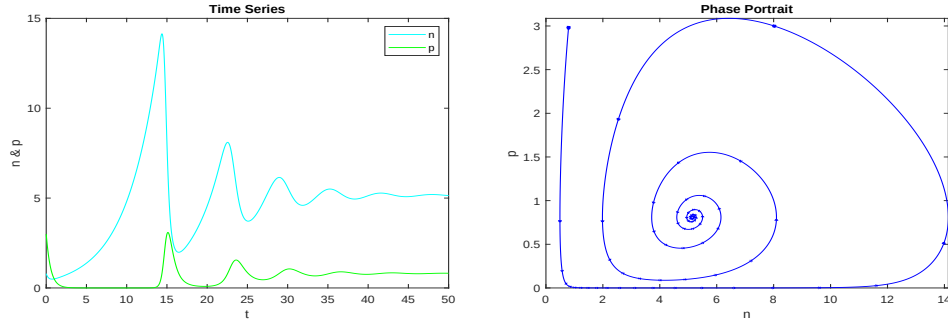


Figure 5.1: Time series and Phase Portrait in the case where $B < 0$. Here, we see that the equilibrium point is a stable focus. Parameters have been chosen as: $\delta = 2, \alpha_0 = 1, r = 0.5, a = 3, \bar{b} = 2, M = 2, b = 1$ and $c = 0.5$.

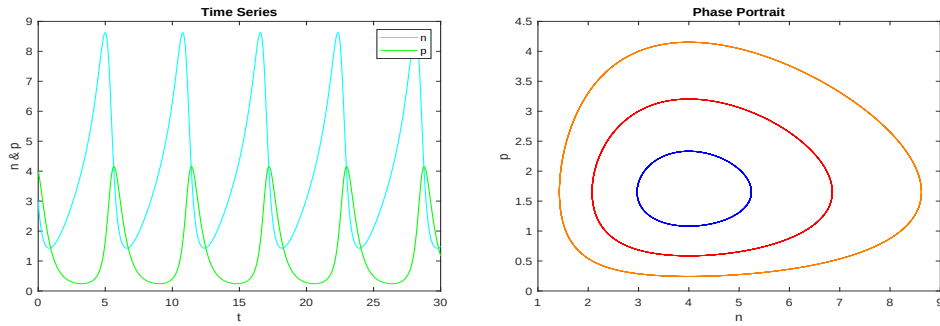


Figure 5.2: Time series and Phase Portrait in the case where $B = 0$. Here, we see that the equilibrium point is a centre. Parameters have been chosen as: $\delta = 2, \alpha_0 = 1, r = 0.5, a = 3, \bar{b} = 2, M = 2, b = 1$ and $c = 1$.

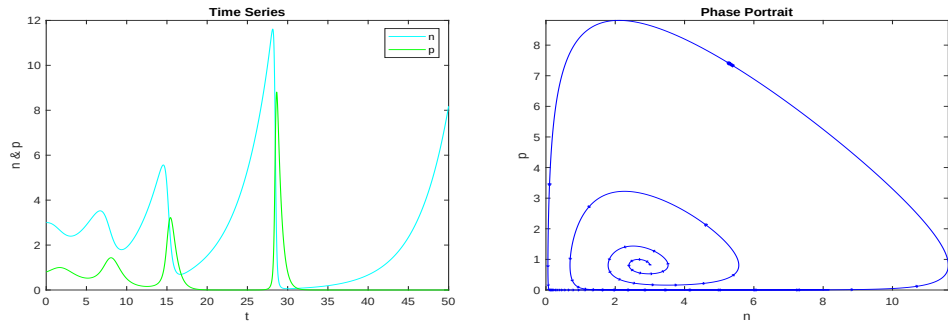


Figure 5.3: Time series and Phase Portrait in the case where $B > 0$. We see clearly that the trajectory tends to the extinction of prey or predators. Parameters have been chosen as: $\delta = 2, \alpha_0 = 1, r = 0.5, a = 1, \bar{b} = 2, M = 2, b = 1$ and $c = 2$,

Chapter 6

Conclusion & Suggestion of Research Project

6.1 Conclusion

The analysis of the aggregated model (5.4) tells us three main situations:

- The non-trivial equilibrium (n^*, p^*) given by (5.7) is stable. In this case, prey and predator coexist with constant densities on each patch.
- The non-trivial equilibrium (n^*, p^*) is unstable. In this case, numerical simulations (Figure 5.3) show that, after some time, the trajectories pass very close to the n and p axes, which would lead, in a real situation, to the extinction of the predator-prey population.
- The non-trivial equilibrium (n^*, p^*) is a center. In this case, Figure 5.2 shows that there are closed trajectories. Predator and prey densities cycle with different amplitudes depending on the initial conditions.

It is useful to compare the dynamics of the local model on each isolated patch to that of the aggregated model. Unlike the local model, when the patches are connected by migrations, the aggregated model comes into play. So, the qualitative dynamics of the single patch and of the two-patch model are different. Consequently, we say that this repulsive effect can promote stability in the case of a stable equilibrium.

6.2 Suggestion

In this study, we have studied the systems for some special conditions as (5.2). However, the case when $m(p_1) = \alpha p_1^h + \alpha_0$, $m(p_2) = \beta p_2^h + \alpha_0$, where $h > 1$, needs to be studied. Also, the general theory of density-dependent migrations that we have studied needs to be properly analyzed (which was not possible to include in this report due to time constraints). One interesting case to study in this regard is a repulsive–attractive model; i.e., one can assume that when the predator density is large on a patch, the prey dispersal rate is large too, and at the same time, one will assume that predators will remain on a given patch when the prey density is large and leave it when it is small, and so on. With advancements in technologies and theoretical analysis, better research work can be done in this discipline.

Chapter 7

Results Achieved

7.1 Stability and Stability Parameter

In this study, we have encountered one parameter $B := \frac{p^* M(2\alpha_0 c - \delta b)}{(b + cp)(\delta p^* + 2\alpha_0)}$, whose changes of values alter the stability of the system. Moreover, the stability depends on $Q = 2\alpha_0 c - \delta b$. The details is given in the following in tabular form:

Value of Q	Stability
$Q < 0$	Stable focus
$Q = 0$	Centre
$Q > 0$	Unstable focus

7.2 Degenerate Bifurcation

When B crosses the zero value, a Hopf bifurcation occurs. With the help of Dulac's criterion and Bendixon's theorem, we showed that this bifurcation is not associated with limit cycle, contrary to the fact that Hopf bifurcation generally comprises of apparition or the disappearance of a limit cycle. That's why, we deduce that the Hopf bifurcation depicted here is degenerate bifurcation.

Chapter 8

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